

Expectations

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: PMFs

Suppose X is a random variable with the following PMF and support:

$$f(x) = c(x^2 + 1) \quad x = \{-1, 0, 1\}$$

What is the value of c ?

- A. $1/2$
- B. $1/3$
- C. $1/5$
- D. None of the above

Mean and variance

Probability distributions have both a **mean** and **variance**.

Mean: the average outcome of a random variable

$$\mu = \sum_{S_x} x f(x)$$

Variance: average “spread” of a random variable around its mean

$$\sigma^2 = \sum_{S_x} (x - \mu)^2 f(x) = \sum_{S_x} x^2 f(x) - \mu^2$$

→ $\sqrt{\sigma^2} = \sigma = \text{standard deviation}$

Expected value

We often refer to the mean as the **expected value** of a variable.

$$E[X] = \mu = \sum_{S_x} x f(x)$$

This formula can apply to any function of X :

$$E[g(X)] = \sum_{S_x} g(x) f(x)$$

E.g., if $g(X) = X^2$ we can compute:

$$E[X^2] = \sum_{S_x} x^2 f(x)$$

E.g., if $g(X) = X^3 + 2X$ we can compute:

$$E[X^3 + 2X] = \sum_{S_x} (x^3 + 2x) f(x)$$

Variance formula

Using the expected value, we can simplify variance:

$$\sigma^2 = E[(X - \mu)^2] = E[(X - E[X])^2]$$

With some algebra, one can show:

$$E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

Properties of expectations

The $E[\cdot]$ operator has some useful rules:

$$E[c] = c$$

$$E[cX] = cE[X]$$

$$E[X \pm Y] = E[X] \pm E[Y]$$

$$E[aX \pm bY] = aE[X] \pm bE[Y]$$

We can also use the $Var(\cdot)$ operator:

$$Var(c) = 0$$

$$Var(aX) = a^2 Var(X)$$

$$Var(aX \pm bY) = a^2 Var(X) + b^2 Var(Y) \quad (\text{if } X \perp Y)$$

Expectations practice

Let X have the following PMF and support:

$$f(x) = \frac{(x^2 + 1)}{5} \quad x = \{-1, 0, 1\}$$

- What is $E[X]$?
- What is $Var[X]$?
- What is $E[X^2 - 2X]$?

Expectations practice

Expectations practice

Consider a 52 card deck, with 4 cards of each rank (J, Q, K, A, etc.). Draw a card at random and define X in the following way: $X = 4$ if an Ace is drawn, $X = 3$ if a King is drawn, $X = 2$ if a Queen is drawn, $X = 1$ if a Jack is drawn, and $X = 0$ for any other card.

- What is $P(X \geq 1)$?
- What is $E[X]$?
- What is $E[X^2]$?

Expectations practice

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Let X have mean 1, and variance 3.

What is the variance of $Y = \frac{X-1}{\sqrt{3}}$?